

Practice Unit Test Solutions – Chemical Systems and Equilibrium

Part A: Multiple Choice

1. D
2. D
3. C
4. A
5. C
6. E
7. C
8. A
9. B
10. D
11. B
12. E
13. C
14. B
15. D

Part B – Fill in the blank 11/15

- a. Homogeneous equilibrium
- b. Heterogeneous equilibrium
- c. resists
- d. Arrhenius
- e. Bronsted Lowry
- f. strong, weak
- g. Le Chatelier's Principle
- h. endpoint, equivalence point
- i.
 - a. decrease concentration
 - b. increase concentration
 - c. increase concentration
 - d. decrease concentration
 - e. no change
 - f. increase concentration
 - g. no change

Part C: Short Answer 10/16

1.

a.

$$K_{sp} = [\text{Fe}^{3+}][\text{OH}^-]^3$$

Since one mole of $\text{Fe}(\text{OH})_3$ yields one mole of Fe^{3+} ions and three moles of OH^- ions in the solution:

$$K_{sp} = (3.40 \times 10^{-6}) \times (3 \times (3.40 \times 10^{-6})^3)$$

$$K_{sp} = (3.40 \times 10^{-6}) \times (3 \times (3.40 \times 10^{-6})^3)$$

$$K_{sp} = 1.63 \times 10^{-34}$$

b.

$$(3.40 \times 10^{-6}) - (0.10/3)$$

$$= 3.40 \times 10^{-6} - 3.33 \times 10^{-2}$$

$$\approx 3.40 \times 10^{-6}$$

2.

Part A:

Let x be the change in moles of the reactants and products at equilibrium.

At equilibrium:

Moles of SO_2 (SO_2): $\text{SO}_2^i - x$

Moles of NO_2 (NO_2): $\text{NO}_2^i - x$

Moles of NO (NO): x

Moles of SO_3 (SO_3): x

The equilibrium expression for the reaction is:

$$K_c = \frac{[\text{NO}][\text{SO}_3]}{[\text{SO}_2][\text{NO}_2]}$$

Substitute the given values:

$$4.8 = \frac{x^2}{(1.7 \times 10^{-1} - x)(1.1 \times 10^{-1} - x)}$$

Since x is assumed to be much smaller than the initial amounts of SO_2 and NO_2 , we can approximate $(1.7 \times 10^{-1} - x)$ and $(1.1 \times 10^{-1} - x)$ as (1.7×10^{-1}) and (1.1×10^{-1}) , respectively.

Now, solve for x :

$$4.8 = \frac{x^2}{(1.7 \times 10^{-1})(1.1 \times 10^{-1})}$$

$$x^2 = 4.8 \times (1.7 \times 10^{-1}) \times (1.1 \times 10^{-1})$$

$$x^2 \approx 0.09168$$

$$x \approx \sqrt{0.09168}$$

$$x \approx 0.303$$

Now, calculate the equilibrium concentrations of all species:

Moles of SO_2 (SO_2) at equilibrium: $\text{SO}_2 i - x \approx 1.7 \times 10^{-1} - 0.303 \approx 1.40 \times 10^{-1} \text{ mol}$

Moles of NO_2 (NO_2) at equilibrium: $\text{NO}_2 i - x \approx 1.1 \times 10^{-1} - 0.303 \approx 0.80 \times 10^{-1} \text{ mol}$

Moles of NO (NO) at equilibrium: $x \approx 0.303 \text{ mol}$

Moles of SO_3 (SO_3) at equilibrium: $x \approx 0.303 \text{ mol}$

Therefore, the equilibrium concentrations of all reactants and products at this temperature are approximately:

$$[\text{SO}_2]_{\text{eq}} \approx 1.40 \times 10^{-1} \text{ mol/L}$$

$$[\text{NO}_2]_{\text{eq}} \approx 0.80 \times 10^{-1} \text{ mol/L}$$

$$[\text{NO}]_{\text{eq}} \approx [\text{SO}_3]_{\text{eq}} \approx 0.303 \text{ mol/L}$$

Part B:

$$Q_c = \frac{(1.50)(3.00)}{(0.20)(1.00)}$$

$$Q_c = \frac{4.50}{0.20}$$

$$Q_c = 22.5$$

a.

$Q_c = 22.5$, $Q_c > K_c$, therefore the reaction proceeds to reactants

$$Q_c = \frac{(7.20)(4.00)}{(1.20)(5.00)}$$

$$Q_c = \frac{28.8}{6.00}$$

$$Q_c = 4.8$$

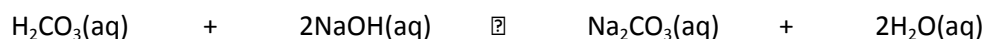
b.

$$Q_c = 4.8,$$

$Q_c = K_c$, therefore at equilibrium

Part D: Long Answer 12/12

Balanced Equation:

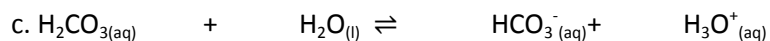
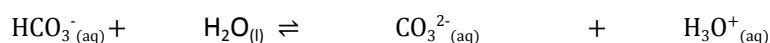
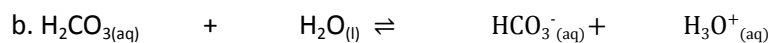


$$1\text{a. } 0.055 \text{ L} \times 1.0 \text{ mol/L} = 0.055 \text{ mol NaOH} \times 1 \text{ mol H}_2\text{CO}_3 / 2 \text{ mol NaOH}$$

$$C = n/v$$

$$= 0.0275 \text{ mol H}_2\text{CO}_3 / 0.025 \text{ L}$$

$$= 1.10 \text{ mol/L}$$



$$\text{I } 1.10 \text{ mol/L} \quad 0 \quad 0$$

$$\text{C } -x \quad +x \quad +x$$

$$\text{E } 1.10 \text{ mol/L} - x \quad x \quad x$$

$$K_{a1} = 4.5 \times 10^{-7}$$

Can ignore (-x) with assumption

$$x = 7.04 \times 10^{-4} \text{ mol/L}$$



$$\text{I } 7.04 \times 10^{-4} \text{ mol/L} \quad 0 \quad 7.04 \times 10^{-4} \text{ mol/L}$$

$$\text{C } -x \quad +x \quad +x$$

$$\text{E } 7.04 \times 10^{-4} \text{ mol/L} - x \quad x \quad 7.04 \times 10^{-4} \text{ mol/L} + x$$

Can ignore (-x) & (+x) with assumption

The $7.04 \times 10^{-4} \text{ mol/L}$ terms cancel out leaving you with

$$K_{a2} = x \quad \text{or} \quad 4.7 \times 10^{-11} = x$$

Since the value of x is so small in the second dissociation reaction it can be assumed that the pH and concentration of $[\text{HCO}_3^-]$ are based on the value of x from the first dissociation reaction. Therefore,

$$[\text{HCO}_3^-] = 7.04 \times 10^{-4} \text{ mol/L}$$

$$\text{pH} = -\log [7.04 \times 10^{-4} \text{ mol/L}] = 3.15$$

$$\text{d. } 7.04 \times 10^{-4} \text{ mol/L} / 1.1 \text{ mol/L} \times 100 = 0.064 \%$$

e. Strong acids completely dissociate in water (100%). From this acid we are only getting 0.032 % dissociation which means the equilibrium lies more towards the reactants.